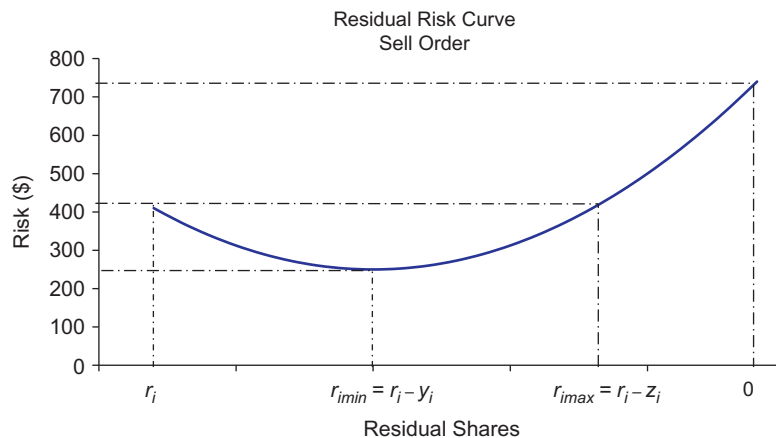


the portfolio and increasing the holding size. However, if a trader needs to adhere to a shrinking portfolio constraint they can only reduce portfolio risk if this minimum value is less than the current number of shares held in the basket.

Figure 9.4 depicts the scenario where the residual risk curve achieves its minimum value. In this example, an investor with a basket of stock has r_i shares of stock i . The figure shows two interesting trading values. The first is $r_{imin} = r_i - y_i$ and represents the number of shares that can be traded to achieve minimum portfolio risk. The second value is $r_{imax} = r_i - z_i$, which represents the maximum number of shares that can be transacted without adversely affecting risk. That is, the residual risk will be exactly the same after trading the shares as it was before trading the shares. These two values will be referred to as: (1) minimum trading risk quantity, and (2) maximum trading opportunity, respectively.

The minimum trading risk quantity is useful for investors continuously striving to take advantage of favorable liquidity conditions to minimize portfolio risk over time. The maximum trading opportunity is useful for investors striving to reduce trading cost without adversely affecting the overall risk of the trade basket. In both situations investors can accelerate transactions in a stock without adversely affecting risk.



■ Figure 9.4 Residual Risk Curve for a Sell Order.